Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

Load Dataset

```
df = pd.read_csv('/content/temperatures.csv')
df.head()
```

	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JAN-FEB	MAR-MAY	JUN-SEP	OCT-DEC
0	1901	22.40	24.14	29.07	31.91	33.41	33.18	31.21	30.39	30.47	29.97	27.31	24.49	28.96	23.27	31.46	31.27	27.25
1	1902	24.93	26.58	29.77	31.78	33.73	32.91	30.92	30.73	29.80	29.12	26.31	24.04	29.22	25.75	31.76	31.09	26.49
2	1903	23.44	25.03	27.83	31.39	32.91	33.00	31.34	29.98	29.85	29.04	26.08	23.65	28.47	24.24	30.71	30.92	26.26
3	1904	22.50	24.73	28.21	32.02	32.64	32.07	30.36	30.09	30.04	29.20	26.36	23.63	28.49	23.62	30.95	30.66	26.40
4	1905	22.00	22.83	26.68	30.01	33.32	33.25	31.44	30.68	30.12	30.67	27.52	23.82	28.30	22.25	30.00	31.33	26.57

Understand Dataset

```
print(df.columns)
print(df.info())
```

```
Index(['YEAR', 'JAN', 'FEB', 'MAR', 'APR', 'MAY', 'JUN', 'JUL', 'AUG', 'SEP',
       'OCT', 'NOV', 'DEC', 'ANNUAL', 'JAN-FEB', 'MAR-MAY', 'JUN-SEP',
       'OCT-DEC'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 117 entries, 0 to 116
Data columns (total 18 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   YEAR     117 non-null    int64
 1   JAN      117 non-null    float64
 2   FEB      117 non-null    float64
 3   MAR      117 non-null    float64
 4   APR      117 non-null    float64
 5   MAY      117 non-null    float64
 6   JUN      117 non-null    float64
 7   JUL      117 non-null    float64
 8   AUG      117 non-null    float64
 9   SEP      117 non-null    float64
 10  OCT      117 non-null    float64
 11  NOV      117 non-null    float64
 12  DEC      117 non-null    float64
 13  ANNUAL   117 non-null    float64
 14  JAN-FEB  117 non-null    float64
 15  MAR-MAY  117 non-null    float64
 16  JUN-SEP  117 non-null    float64
 17  OCT-DEC  117 non-null    float64
dtypes: float64(17), int64(1)
memory usage: 16.6 KB
None
```

Convert Data We convert data into Month vs Temperature format

```
# Convert wide format → long format
df_long = df.melt(id_vars=['YEAR'],
                  var_name='Month',
                  value_name='Temperature')

df_long.head()
```

	YEAR	Month	Temperature
0	1901	JAN	22.40
1	1902	JAN	24.93
2	1903	JAN	23.44
3	1904	JAN	22.50
4	1905	JAN	22.00

Convert Month Names to Numbers

```
month_map = {
    'JAN':1, 'FEB':2, 'MAR':3, 'APR':4, 'MAY':5, 'JUN':6,
    'JUL':7, 'AUG':8, 'SEP':9, 'OCT':10, 'NOV':11, 'DEC':12
}

df_long['Month'] = df_long['Month'].map(month_map)
```

Remove Missing Values

```
df_long = df_long.dropna()
```

Define Features & Target

```
X = df_long[['Month']]   # Input
y = df_long['Temperature']   # Output
```

Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

Apply Linear Regression

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

Predictions

```
y_pred = model.predict(X_test)
```

Evaluation Metrics

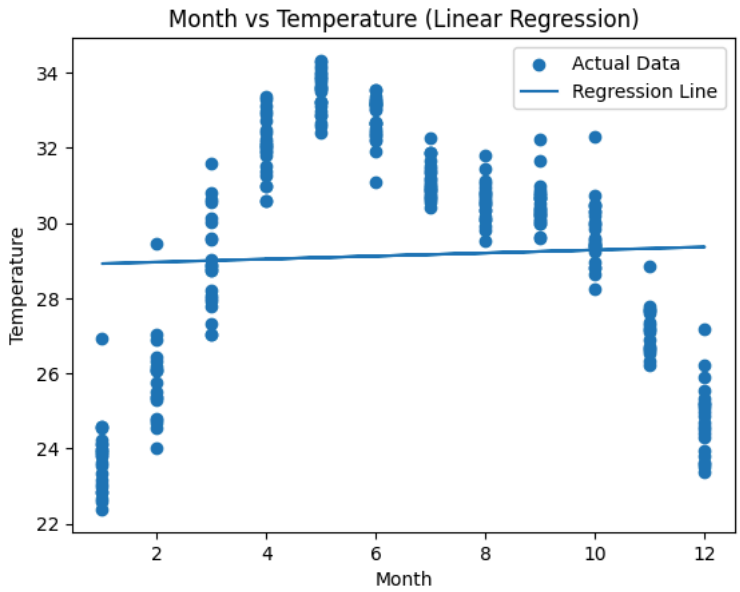
```
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MSE:", mse)
print("MAE:", mae)
print("R2 Score:", r2)
```

```
MSE: 9.999806682283973
MAE: 2.705271869781447
R2 Score: -0.005573628993982727
```

Visualization

```
plt.scatter(X_test, y_test, label="Actual Data")
plt.plot(X_test, y_pred, label="Regression Line")
plt.xlabel("Month")
plt.ylabel("Temperature")
plt.title("Month vs Temperature (Linear Regression)")
plt.legend()
plt.show()
```



Predict Temperature for Any Month

```
# Example: Predict for May (5)
pred = model.predict([[5]])
print("Predicted Temperature for Month 5:", pred[0])
```